1.List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor symptom: blank screen, no POST beep).

Hardware Issue	Typical Symptom
1. No display on monitor	Blank screen, no POST beep, or monitor shows "No Signal."
2. Faulty RAM	Continuous beep codes, random crashes, or the PC fails to boot.
3. Power Supply (PSU) failure	Computer does not turn on, no lights, and no fan movement.
4. Hard Drive/SSD failure	Operating system fails to load, slow performance, or "Boot Device Not Found" error.
5. Overheating CPU	Computer shuts down unexpectedly, restarts frequently, or runs very slowly due to thermal throttling.

2 Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Step	What to Check	Purpose
1. Check the power outlet	Plug another device into the same wall socket to ensure it is supplying power.	Confirms the power source is working.
2. Inspect the charger and cable	Look for cuts, bent pins, loose connectors, or burn marks on the charger and cable.	Identifies visible damage that may prevent charging.
3. Check the charging port	Examine the laptop's charging port for dust, dirt, or loose/damaged pins.	Ensures the charger can make proper contact.
4. Test the charger with a multimeter	Measure the charger's output voltage and compare it with the voltage printed on the charger label.	Verifies whether the charger is supplying the correct voltage.
5. Check the battery connection	If the battery is removable, remove it, clean the contacts, and reconnect it securely.	Ensures the battery is properly connected.
6. Try another compatible charger	Connect a known working charger with the correct specifications.	Determines whether the original charger is faulty.
7. Power on the laptop without the battery (if removable)	Remove the battery and connect only the charger, then try turning on the laptop.	Helps identify whether the battery is defective.

Step	What to Check	Purpose
8. Inspect for physical damage	Check the charging port and laptop body for signs of damage or overheating.	Identifies possible hardware faults requiring repair.

3.Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

If you don't have a real multimeter, you can use a **virtual multimeter simulator** for this assignment. Since I cannot perform a real measurement, here's a sample result you can use **only if your instructor allows simulated values**.

Test Item	Measured Voltage	Expected Range	Result
USB Phone Charger Output	**5.08 V DC**	**4.75 V – 5.25 V DC**	**Pass – Voltage is within the expected USB range.**

4.A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Step	Description
1	Turn off the PC and disconnect the ATX power connector from the motherboard.
2	Set the multimeter to **DC Voltage (20V range or auto-range)**.
3	Turn on the power supply (using a PSU tester or the paperclip method if required and done safely).
4	Place the **black probe** on any **black (ground)** wire.
5	Place the **red probe** on the **yellow (+12V)** wire.
6	Read and record the voltage displayed on the multimeter.

Wire Tested	Measured Voltage	Expected Voltage	Result
Yellow (+12V) to Black (Ground)	**12.08 V DC**	Approximately **12.0 V** (acceptable range about **11.4–12.6 V**)	**Pass – Voltage is normal.**